Mehak Chopra

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Award-winning scientific communicator and collaborative computational biologist with expertise in cardiovascular genomics, large-scale sequencing analysis, and reproducible data science. Experienced in integrating molecular, imaging, and clinical datasets to explore mechanisms of vascular disease.

EDUCATION

University of Galway — **Galway, Ireland**
Ph.D. Genomic Data Science — *September 2022–Present*

Pondicherry University — **Puducherry, India**
M.Sc. Bioinformatics — *July 2019–July 2021*

Panjab University — **Chandigarh, India**
B.Sc. Bioinformatics — *July 2016–May 2019*

RESEARCH EXPERIENCE

University of Galway — **Galway, Ireland**
PhD Research Candidate — *September 2022–August 2026*

Research focuses on application of genomics data analysis as well as bulk and single nucleus RNA sequencing to elucidate the mechanisms of genetic and age-related variation in a clinically relevant aortic phenotype and aortic aneurysms.

University of Liverpool — **Liverpool, UK**
Visiting Research Fellow — *March 2025*

Gained practical experience working with biobanked aortic tissue samples from diseased aorta, involving exposure to collection procedures and an understanding of how biomechanical properties of aortic tissues are measured and analyzed for research applications.

Yale University — **New Haven, USA**
Visiting Postgraduate Fellow — *October 2024–January 2025*

Collaborated closely with clinicians to deepen understanding of cardiovascular diseases while applying advanced statistical and bioinformatic methods to analyze gene expression and whole-exome sequencing data, leading to the identification of key genomic insights.

Centre for Brain Research - Indian Institute of Science — **Bangalore, India**
Data Analyst — *October 2021–August 2022*

Managed and analyzed genotypic data from the Indian population as part of the *Genome India Project*, developing a copy number variation detection pipeline and conducting evolutionary genomics analyses to trace gene flow and evolutionary links between ancient and modern human populations.

Pondicherry University — **Puducherry, India**
MSc Project — *February 2021–June 2021*

Undertook a Master's research project titled *Triad of Coronary Heart Disease, Hyperlipidemia, and Diabetes*, aiming to identify shared molecular signatures among these conditions through differential gene expression analysis of RNA-seq data, complemented by gene ontology, enrichment, and pathway analyses.

Open Life Science - OLS
Online Research Project — *February 2021–May 2021*

Completed a group project titled *Systems Genomics Integration of Diabetes-Related Genes: A Quest for Development of Biomarkers,* aimed at identifying common genes shared among type 1 and type 2 diabetes and obesity using systems biology approaches, and exploring correlations between the human gut microbiome and these genes to identify potential candidates for rational drug design.

Panjab University **Chandigarh, India**
Bsc Major Project *January 2019–April 2019*

Collaborated with research peers to develop machine learning models using LIBSVM for alkaloid prediction using GAUSSIAN-optimized molecular structures, PubChem-derived physicochemical descriptors.

Panjab University **Chandigarh, India**
BSc Minor Project *January 2018–April 2018*

Undertook a Bachelor's project titled *In Silico Mutational Analysis of Interleukin-10*, which involved using bioinformatics approaches to induce mutations in protein structures and assess the stability differences between mutant and wild-type proteins through receptor docking analysis.

PUBLICATIONS

- **Chopra M**, Hynes N, Seoighe C. Impact of cell type specific variations and age in aortic distensibility. medRxiv. 2025 Nov 27:2025-11. doi: 10.1101/2025.11.25.25340959. *(preprint)*
- Seoighe C, Connaire S, **Chopra M**. Probing the limits of cis-acting gene regulation using a model of allelic imbalance quantitative trait loci. PLoS genetics. 2025 Apr 30;21(4):e1011446. doi: 10.1371/journal.pgen.1011446.
- Subramanian K, **Chopra M**, Kahali B. Landscape of genomic structural variations in Indian population-based cohorts: deeper insights into their prevalence and clinical relevance. Human Genetics and Genomics Advances. 2024 Jul 18;5(3). doi: 10.1016/j.xhgg.2024.100285.
- Kaur A, **Chopra M**, Bhushan M, Gupta S, Kumari P H, Sivagurunathan N, Shukla N, Rajagopal S, Bhalothia P, Sharma P, Naravula J. The omic insights on unfolding saga of COVID-19. Frontiers in immunology. 2021 Oct 20;12:724914. doi: 10.3389/fimmu.2021.724914.

TEACHING EXPERIENCE

Teaching Assistant **University of Galway**
M.Sc. in Genomic Data Science *September 2023 - December 2025*

As a teaching assistant, I prepared and presented material for practical sessions, led training sessions, and provided feedback to the teaching team based on submission grades for the following modules:

- MA5116 - Introduction to Probability for Genomics
- MA215 - Mathematical Molecular Biology I
- MA216 - Mathematical Molecular Biology II
- MA5106 - Medical Genomics I

GRANTS

Open Targets Hackathon Funding **EMBL**
Lead Applicant *October 2025*

- Secured sponsorship to attend the Open Targets Hackathon; collaborated on a project that received the Best Project Award.

Research Mobility Grant **European Ageing Network**
Lead Applicant *March 2025*

- Funded a one-month research visit to the University of Liverpool to collaborate with the LABB (Liverpool Aortic Biomechanics and Biochemistry Research) group; gained hands-on experience in assessing aortic biomechanical properties.

RESEARCH COMPLIANCE

April, 2024: Prepared and submitted ethics application for a study on molecular and biomechanical properties of healthy and diseased aortic tissue; successfully obtained approval from the HSE Galway Ethics Research Committee, enabling sample collection and pilot study execution.

February, 2025: Organized and conducted mini-competition among vendors in line with University of Galway's procurement regulations, ensuring transparency and compliance.

SELECTED PRESENTATIONS

Delivered oral and poster presentations on PhD research exploring genomic and cellular determinants of aortic distensibility at multiple national and international meetings.
Presented at Surgical Research Society Annual Meeting 2026 (London, UK); VIBE 2025 (Dublin, Ireland); EuroAgeNet "Cell Communication Networks in Ageing and Disease" Conference 2025 (Liverpool, UK); Aorta Meeting 2024 (Liverpool, UK); Indian Society of Human Genetics 2024 (Gujarat, India); Irish Computational Biology and Genomics Symposium 2023 (Galway, Ireland); and University of Galway Research Day 2025.

AWARDS AND PRIZES

- **Best Poster Award, EuroAgeNet Conference 2025 (UK).** Recognized for presenting research on cell type specific gene interactions in aortic distensibility.
- **Best Lightning Talk, School of Mathematical and Statistical Sciences Research Day 2025 (University of Galway).** Awarded for an engaging and accessible presentation related to PhD research work.
- **Student Scholar Award, Indian Society of Human Genetics 2024 (India).** Received travel support for scientific merit in abstract submission.
- **Best Presentation, Irish Computational Biology and Genomics Symposium 2023 (Ireland).** Selected as the top oral presentation among early-career researchers.
- **CRT Travel Fellowship, ASFHG African Genomics Meeting 2023 (South Africa).** Fully funded award to present a selected group proposal on African population genomics.
- **Best Project Award, Open Targets Hackathon 2025 (UK).** Secured full sponsorship to attend and collaborated on a computational genomics project (GitHub: `gwasXsinglecell_OT_hackathon`).
- **Best Oral Presentation, Indian conference of bioinformatics 2019 (India).** Recognized for presenting undergraduate research work.

POSITIONS OF RESPONSIBILITY

- One of two student representatives for my PhD cohort (2022–2026), which involved planning events and liaising with the program managers if any students had issues.
- Served on the PhD Programme Governance Committee (2023–2026), contributing student perspectives to institutional decisions and policy discussions.
- Mentored an undergraduate student on a research project (2024), providing guidance in data analysis.
- Member of the Organising Committee, Virtual Institute of Bioinformatics and Evolution/Irish Computational Biology and Genomics (VIBE/ICBG) Symposium (2023 and 2024).

REFERENCES

Prof. Cathal Seoighe
Professor of Bioinformatics and Scientific Director
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Dr. Niamh Hynes
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